Results

Tests Performed

After each model was trained, various checkpoints from each model were evaluated using 6 out of distribution novel test views.

The following metrics were used:

- **SSIM (Structural Similarity Index Measure):** Compares the original and reconstructed image based on perceived structural information (luminance, contrast, structure). *Higher is better*
- **PSNR (Peak Signal-to-Noise Ratio):** Compares the original and reconstructed image pixel by pixel. *Higher is better.*
- **LPIPS (Learned Perceptual Image Patch Similarity)** [1]**:** Uses a deep neural network trained to predict human perceptual similarity judgments. *Lower is better.*

Quantitative Results

Model Configuration	Iterations	Test SSIM	Test PSNR	Test LPIPS
Sparse Views + Refinement	45k	0.774	25.9	0.271
Sparse Views, No Refinement	15k	0.395	18.53	0.404
Sparse Views, No Refinement	30k	0.381	18.81	0.418
Sparse Views, No Refinement	45k	0.522	18.37	0.353
Dense Views	15k	0.601	21.25	0.216
Dense Views	30k	0.597	21.15	0.215
Dense Views	45k	0.597	21.12	0.215

The table above shows the performance metrics. Note that the 'Sparse Views + Refinement' model row reflects results after 30,000 standard iterations followed by 15,000 iterations of generative refinement (total 45,000 iterations).

Quantitative Analysis

Refinement is Effective for Sparse Views

Comparing the 'Sparse Views + Refinement' model against the 'Sparse Views, No Refinement' model shows a dramatic improvement across all metrics (SSIM, PSNR , LPIPS) when refinement is used. This validates the inclusion of a refinement step for improving representation quality with sparse input views.

Other Comments

The Refined Sparse Model can Outperform Dense Model

Comparing the best sparse result to the best dense result, refined sparse model achieves higher SSIM and higher PSNR. I believe the refined sparse model likely surpasses the dense model on SSIM/PSNR because its refinement process can interpolate or synthesize views geometrically closer to the specific test poses than any single view in the discrete dense set. This refinement can also act as a form of regularization, removing sparse-view artifacts and promoting smoother reconstructions that align well with the structural and pixel-level comparisons. I do note that the dense model still achieves a better LPIPS score, suggesting it might retain better perceptual similarity.

Dense Views Model Converges Quickly

Results for the 'Dense Views' at all iterations are similar across all metrics; further training yields little benefits.

Qualitative Results

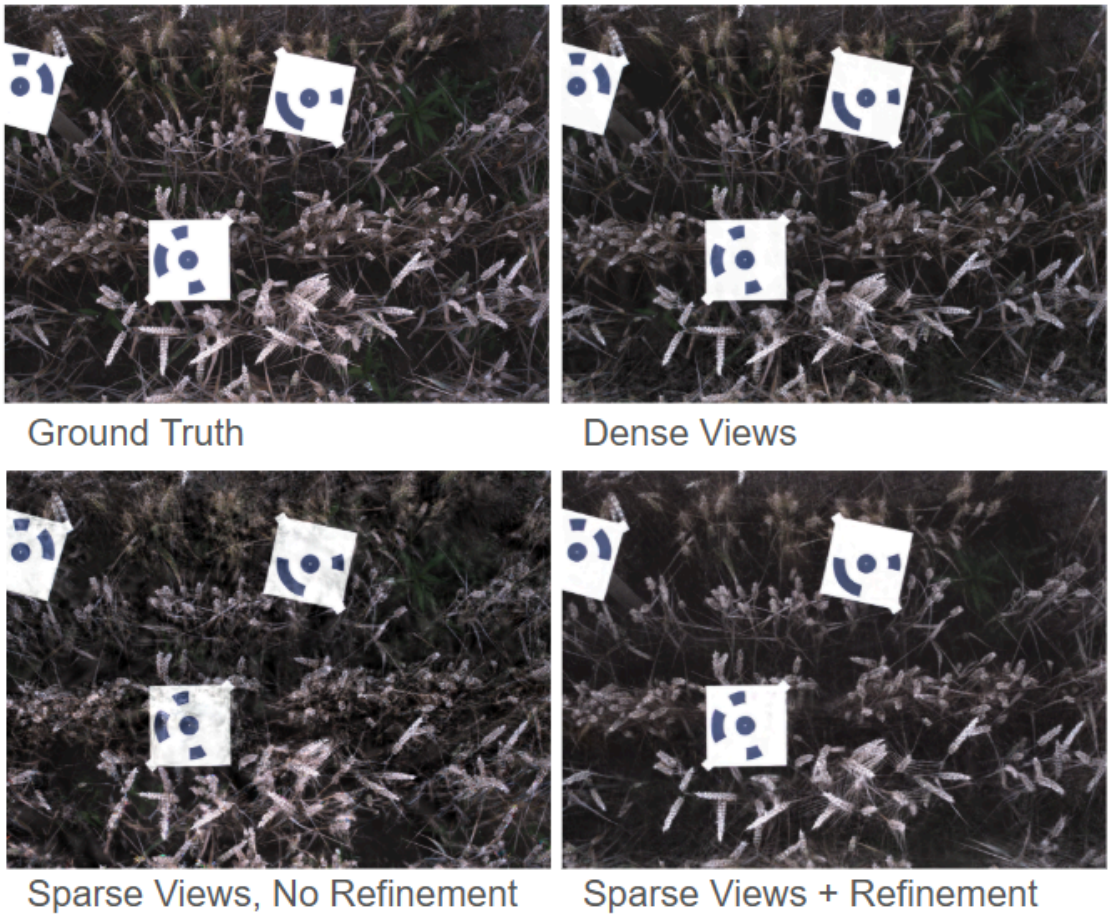


Figure 1: Comparison at 45k Iterations

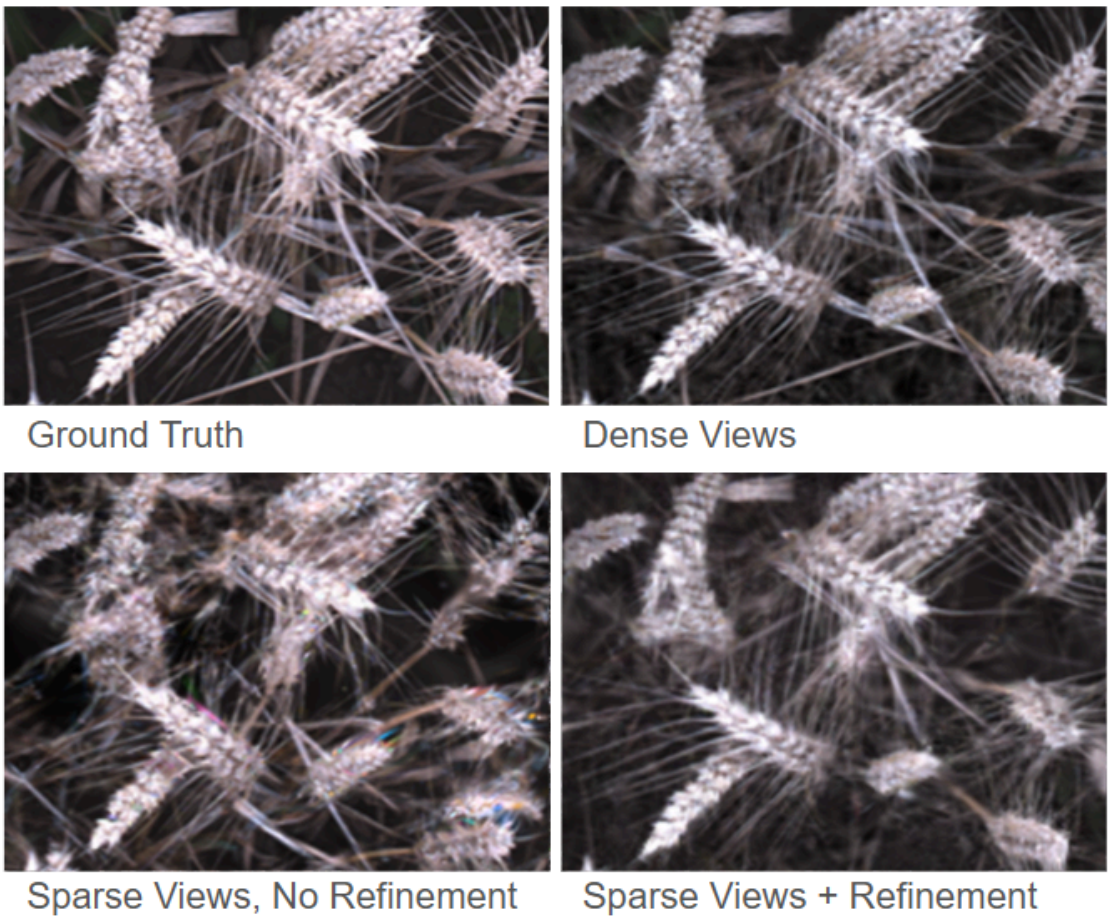


Figure 2: Close Up Comparison at 45k Iterations



Video 1: Rendered comparison across viewpoints.

Qualitative Analysis

Artifact Reduction

Refinement reduces noise, blur, and fragmentation artifacts inherent in the sparse baseline. In Figure 2, the refined model shows clearer edges and structure, appearing smoother overall compared to the 'Sparse Views, No Refinement' model. Visually, the 'Sparse Views + Refinement' model looks most similar to the 'Dense Views' model.

Clarity

Video 1 demonstrates that refinement significantly boosts clarity and temporal coherence. It eliminates the blur and noise of the unrefined sparse model, resulting in a clear rendering across views.

---

1. Zhang, Richard, et al. "The unreasonable effectiveness of deep features as a perceptual metric." *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*. 2018. ↩